EEE-452: ANTENNA ENGINEERING PROJECT PROPOSAL

YAGI UDA ANTENNA DESIGN

Tuna Şahin (22201730) | Berkay Özer (22202119) | Zeynep Nalçakan (22203100)

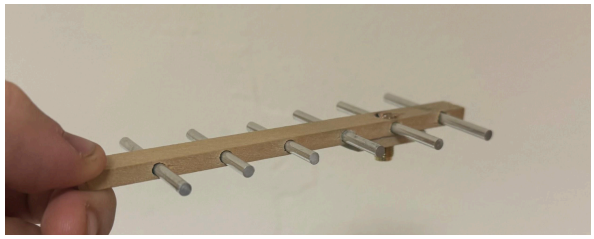

Figure 1: Antenna's Hardware Implementation

1. Recap of the Project:

As a team, we proposed implementing a Yagi-Uda antenna for our term project, as it would be easier to access recycled materials. Additionally, we considered that tuning the antenna can be achieved by adjusting the length of the conducting rods, which allows us to make adjustments without redesigning the antenna from scratch, which made this choice viable. Our group researched and simulated two different types of Yagi-Uda antennas: conventional and quasi-Yagi-Uda antennas. Comparing the S_{11} values, we believed it would be easier to match the conventional Yagi-Uda, as it has a much lower S_{11} value and features a narrower bandwidth. We expected the quality factor to drop in the hardware implementation due to the lossy nature of the components, which will result in a wider band and a lower S_{11} value. The conventional Yagi-Uda design offers a greater margin for error. Therefore, in the hardware implementation, we decided to implement a conventional Yagi-Uda.

Manufacturing Process and Bill of Materials (BOM):

- A short wooden stick was repurposed from scraps of old projects and used as the boom of the antenna.
- For the conductive elements, an aluminum rod with a diameter of roughly 3mm was used, which was obtained from broken camping tents.
- The SMA connectors were provided by Ufuk Tufan at EE-102, Bilkent.

Since all of the materials were acquired from scraps, the project cost is 0₺.

The dimensions of these newly acquired items (i.e., the radius of the aluminum rod and the dimensions of the boom) were entered into CST to compensate for the changes. The dimensions of the remaining components were scaled by a factor of 0.96. The diameter of the elements was reduced from 4 to 3. The elements were cut using a handsaw and inserted into holes on the boom, which were cut using a hand drill. The antenna is visible in Figures 1 and 2.2.

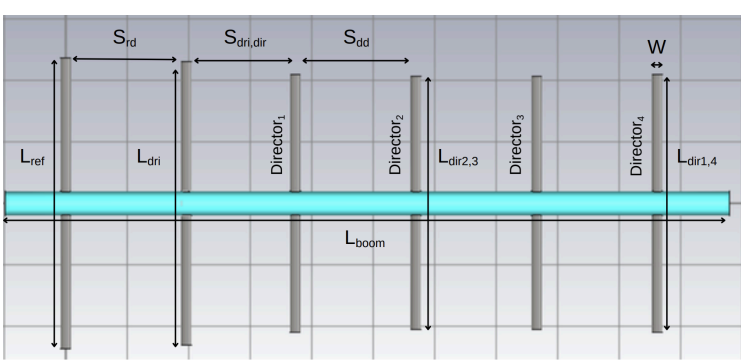

Figure 2.1: Antenna's Design

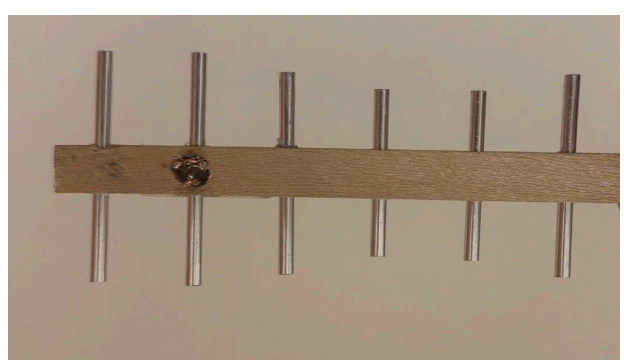

Figure 2.2: Antenna's Hardware Implementation

$\lambda = 94mm$	$\epsilon_{r,boom} = 1.74$	$W_{ref} = W_{dir} = W_{dri} = 3\text{mm}$		$L_{boom} = 1.1\lambda$	$L_{dir2,3} = 0.420\lambda$
$S_{rd} = 0.2\lambda$	$S_{dir,dir} = 0.2\lambda$	$L_{dir1,4} = 0.428\lambda$	$L_{dri} = 0.471\lambda$	$L_{ref} = 0.482\lambda$	$S_{dri,dir} = 0.18\lambda$

Table 1: Dimensions of the antenna

	Simulation	Physical Implementation
Boom	Wood $\epsilon_r = 1.74$	Wood
Conductive Elements	Aluminium	Aluminium

Table 2: Materials of Antennas

2. Further Simulation Results:

The E and H planes were determined. The H-plane is the plane bisecting the antenna. The E-plane is the plane where the elements are located.

The cross-polarization and co-polarization of the antenna were also measured in addition to the results in the analysis report.

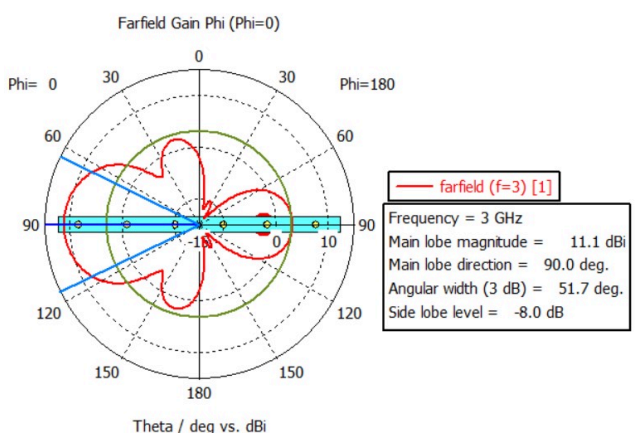

Figure 2.1.1: Co-Polarization (H plane)

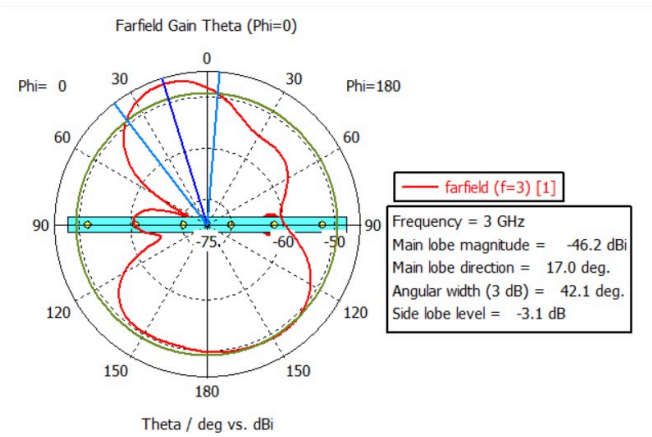

Figure 2.1.2: Cross-Polarization (H plane)

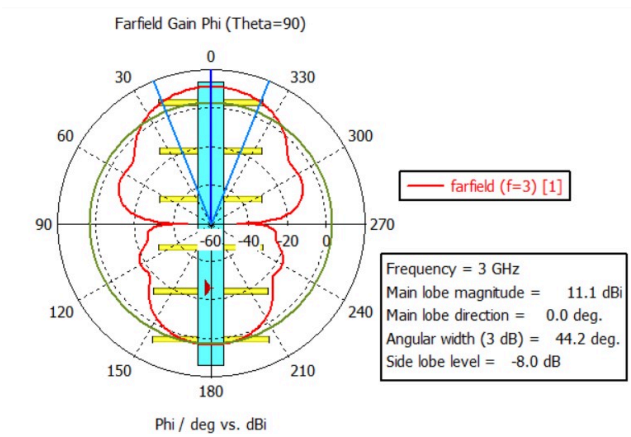

Figure 2.2.1: Co-Polarization (E plane)

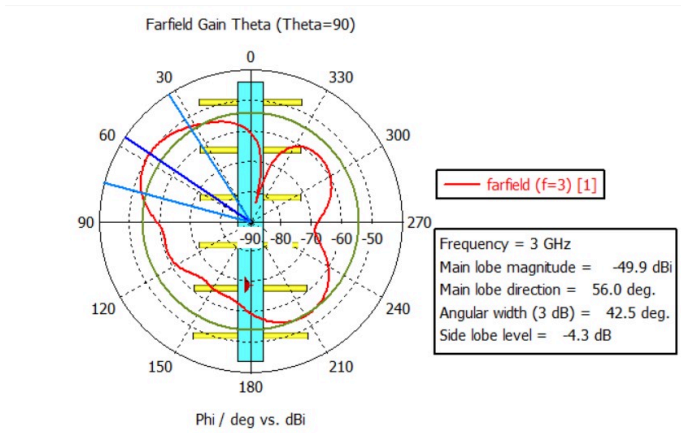

Figure 2.2.2: Cross-Polarization (E plane)

3. Experimental Results and Procedure:

The matching was adjusted by scaling the elements and changing the dielectric constant of the boom. The resulting S_{11} results were obtained after tuning the antenna to 3GHz. For the S_{11} measurements, the VNA in EEE 315 was used. The calibration of the network analyzer was achieved by opening the file:

"D:\sobhan\300KHZ_4GHZ_10KHZ_2PORT_ECAL.-S11=S21STA.STA"

Then the S_{11} measurement was obtained, and when compared with the simulation results, it is observed that the S_{11} got broader due to the increasing loss in the hardware implementation, which decreases the quality factor (Q), as expected.

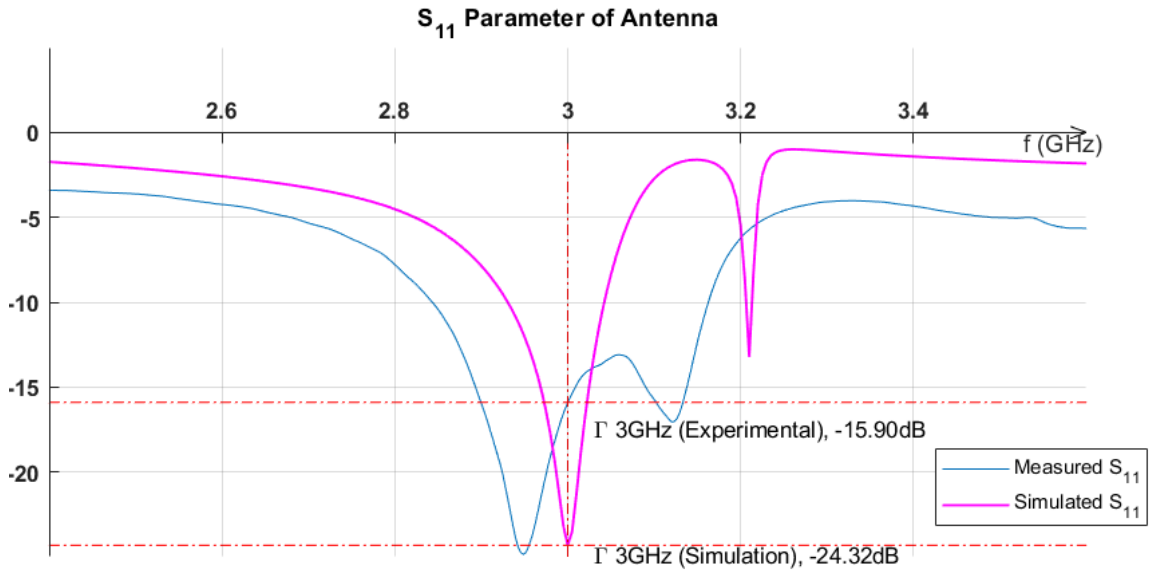

Figure 3.1: Experimental S_{11} Measurements

After the impedance measurement, the antenna's pattern was measured at METEKSAN Defence's facilities. The measurements were performed in an anechoic chamber using near-field measurement methods. AUT was used as a receiver antenna, and as a transmitting antenna, a wide-band (1-20 GHz) ridged horn antenna was used. The system was calibrated using a horn antenna whose gain is known, and the gain of the measured antennas was determined by comparing the amplitude with that of the calibration horn antenna (gain comparison method). The distance between the antennas was set to be 1m, and the AUT was mounted on a plastic bracket on a rotator. Pattern measurements were completed in approximately 12 minutes. The following measurements were obtained.

	E-Plane HBPW (degrees)	H-Plane HBPW (degrees)	Maximum Directivity (dBi)	Radiation Efficiency (dB)	Θ_{max} (degrees)	ϕ_{max} (degrees)	SLL E-Plane (dB)	SLL H-plane (dB)
Experimental 2.9GHz	33.75	51.32	12.651	-3.5	-11	-0.74	-11	-8.5
Experimental 3GHz	52.73	52.73	11.49	-3	-11	-1	-9.6	-9
Experimental 3.1GHz	46.40	57.65	11.51	-3.9	-10	0.7	-7.6	NaN
Simulated 3GHz	44.2	51.8	11.6	-0.5	0	0	-9.6	-9
Theoretical Expectations	30-55	40-60	10-11	~	0	0	~	~

The simulation used lossy materials to make the simulation as close to the physical implementation as possible.

Table 3: Simulated and Measured Antenna Parameters

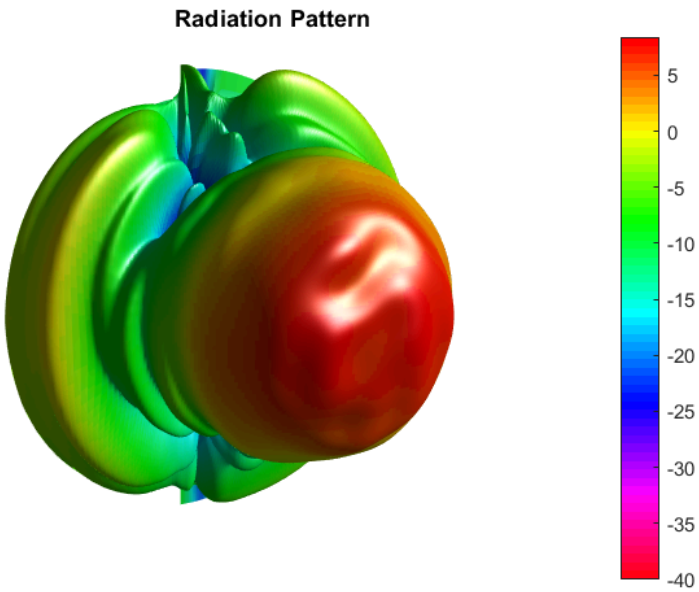

Figure 3.2.1: Measurement of Antenna's 3D Pattern at 3GHz

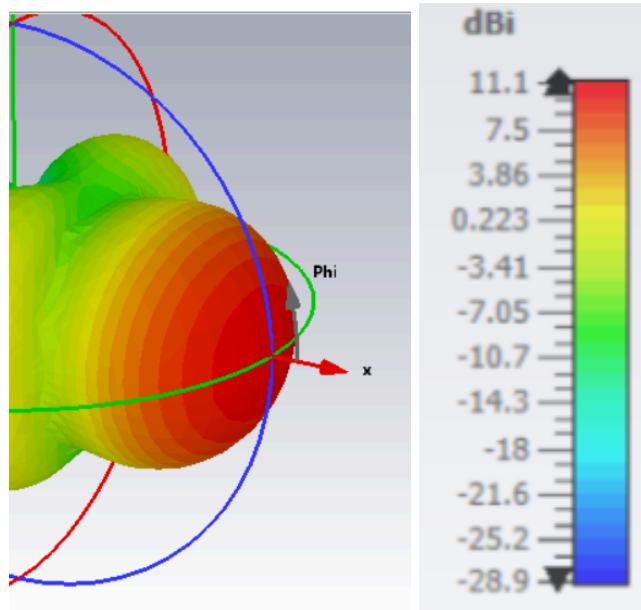

Figure 3.2.2: Simulation of Antenna's 3D Pattern at 3GHz

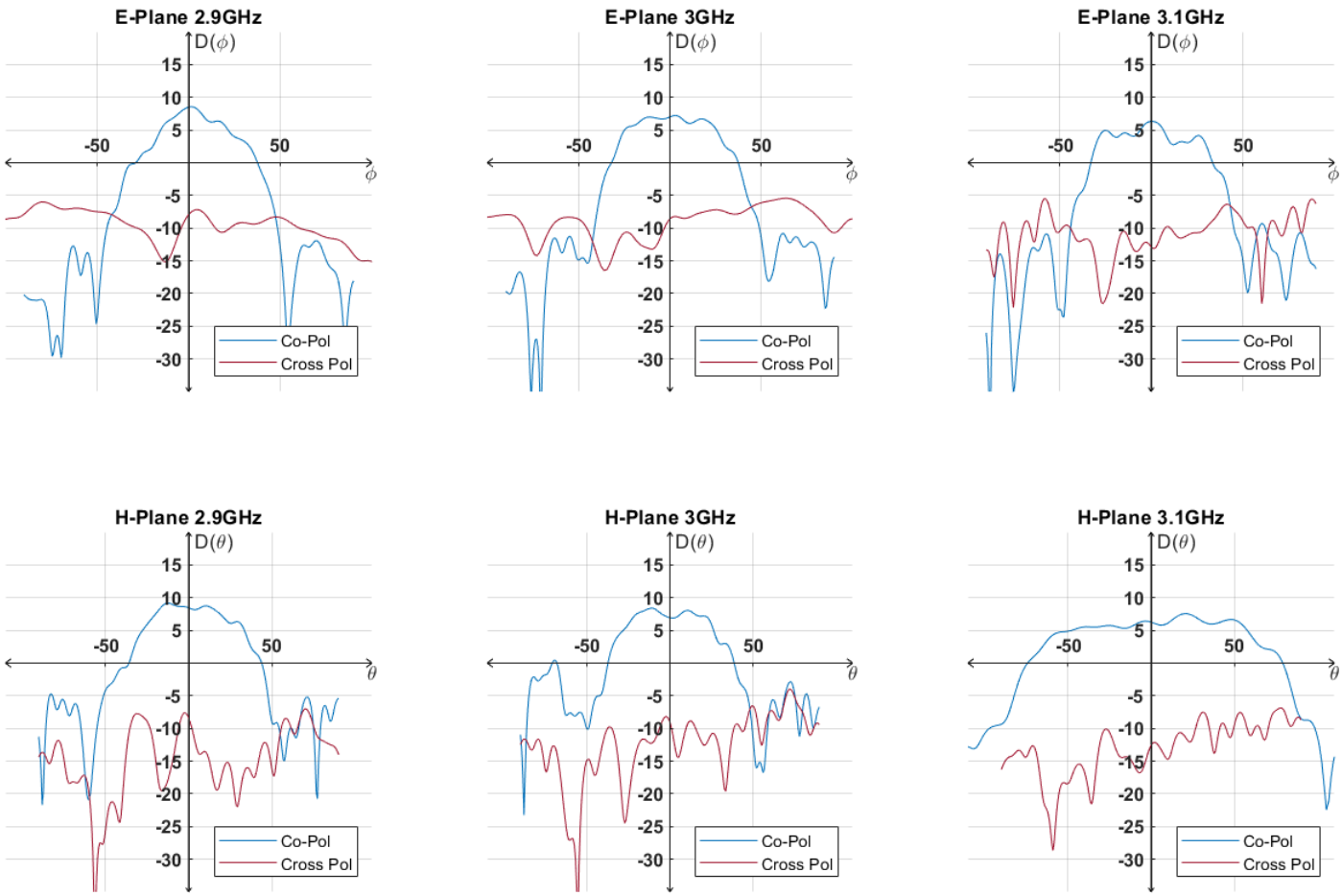

Figure 3.3: Cross and Co-Polarization Gain Patterns for Different Frequencies

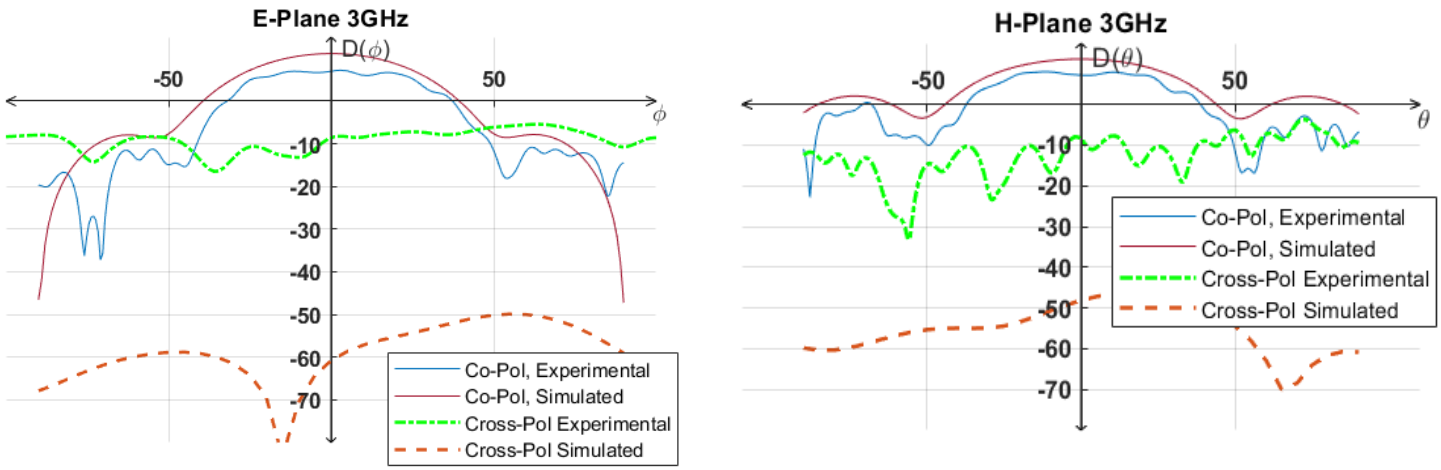

Figure 3.4: Cross and Co-Polarization Comparison at 3GHz

4. Conclusion

- The experimental results generally align with the simulation results. Values for SLL, HPBW, directivity, co-polarized pattern, and S_{11} parameter all align with the simulations, with a slight shift in the resonance frequency for the S_{11} parameter.

- The measurement only measured azimuth values in the range of [-90, 90]. This angle covers half of the solid angle 4π, which underestimates the value of P_{rad} when calculating the directivity. Therefore, this leads to higher directivity and worse radiation efficiency, which account for the discrepancy between directivity and gain. This effect is especially present when the AUT features a significant backlobe, which is the case for our Yagi-Uda. So the actual directivity is lower, and the efficiency is higher than the measured value.

- The direction of maximum radiation was shifted by 10° in the elevation plane. This is due to the dielectrics used to mount the antenna and the additional wooden piece that is used to mechanically reinforce the feeding network.

- Radiation efficiency dropped significantly compared to the simulation, even when the simulation used lossy components. This drop may be due to feedline loss or loss at the connection between the dipole and the SMA connector.

- The measured cross-polarized pattern was much higher than the simulated pattern. One reason for this is the reflections from the backlobe, which causes co-pol leakage, considering the simulated backlobe of the antenna was significant. Also, the finite polarization purity of the receiver antenna will make accurate measurements difficult. Another reason is the possible misalignment of the antennas during installation, which may cause co-polarized leakage. The noise floor will also be significant when measuring values around -70dBi.

REFERENCES:

[1] C. A. Balanis, *Antenna Theory: Analysis and Design*, 2nd ed. New York, NY, USA: John Wiley & Sons, 1997.

[2] Peter P. Viezbicke, *Yagi Antenna Design*, NBS Technical Note 688, December 1976.